CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (CL) Examination

2014

CL305.01 Environmental Engineering-II

Date: 13th May, 2014 Time: 10:00 a.m. To 01:00 p.m.

Day: Tuesday Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Following is a list of units normally used in a sewage treatment plant. Mention for each unit which treatment level it belongs to (i.e. preliminary, primary, or secondary) and what impurity it removes from sewage. Also draw a neat sketch of sewage treatment scheme using listed units. [07]

1. Grit chamber 4. Bar screen 7. Trickling filter
2. Primary clarifier 5. UASB
3. Secondary clarifier 6. Aeration tank

Q - 2 (a) Explain self cleansing and non-scouring velocity in sewers. [03]

(b) Draw a neat graph of BOD v/s incubation period and describe first stage and second stage BOD. [05]

(c) Compute velocity through a coarse screen when approach velocity is 0.6 m/s and the measured head loss is 30 mm. What will be the head loss in this case when the screen is half-clogged? [06]

OR

Q - 2 (a) Briefly describe various limitations of BOD test. [03]

(b) Do as stated: [05]

(i) In the COD test a sample is digested with potassium dichromate in the presence of sulphuric acid. Which of these chemicals (Manganus sulfate, silver sulphate, mercuric sulphate) are added respectively, as a catalyst and a preventer of chloride interference?

(ii) For a sewage sample, BOD value is less than COD value. Why?

(iii) What happens to following parameters when wastewater temperature increases? Also briefly mention the reason. (1) DO (increase/decreases), (2) rate of biodegradation (increase/decreases), (3) velocity of settling of given particles in this wastewater (increase/decreases).

(c) Which type of settling prevails in a primary sedimentation tank used for sewage treatment? Design a circular primary sedimentation tank for a population of 150000. Also calculate SOR and WLR. [06]

Q - 3 **Attempt any Two:** [14]

(a) (i) The BOD of a sewage sample incubated for seven days at 20 °C (BOD_7^{20}) has been found to be 150 mg/L. What will be the 3-day 30 °C BOD? Assume k = 0.23 day^{-1} (base 10) at 20 °C.

(ii) Explain briefly various methods of aeration employed in activated sludge treatment.

(b) Determine size of a circular sewer to serve a population of 300000 persons who are supplied water @ 135 LPCD. Assume that 80% of water gets converted into sewage and peak flow/average flow = 3. The slope available is 1 in 500. Assume that it is flowing 70% full at peak flow. Is the velocity at peak flow self-cleansing? Take N = 0.013.

(c) Design a conventional activated sludge plant to treat 20 MLD of primary settled sewage of BOD = 250 mg/L. The effluent BOD required is 30 mg/L. Use following details: F/M ratio = 0.32 day^{-1}, MLSS = 2500 mg/L, SVI 110 mL/g.

SECTION – II

Q - 4 (a) Differentiate between the working of screen chamber and grit chamber. [02]

(b) Define Primary & Secondary air pollutants and give at least three examples of each. [05]

Q - 5 (a) Identify various sources of the following air pollutants & also state their effects. [03]
i. Sulphur dioxide, ii. Carbon Monoxide, iii. Oxides of Nitrogen

(b) Explain the impacts of improper disposal of solid wastes on human health & environment in general. [04]

(c) What are the general applications of the following air pollution control equipments? [03]
(i) Cyclones (ii) Fabric Filters (iii) Electrostatic precipitators

(d) Write a short note on Landfilling for waste disposal. [04]

OR

Q – 5 (a) Explain the mechanism of action of air pollutants on human beings. [03]

(b) Write a note on incineration of refuse. What are its Advantages & Disadvantages? [04]

(c) Distinguish between Dry & Wet Precipitators. [03]

(d) Draw flow chart indicating functional elements of solid water management. [04]

Q – 6 **Attempt any Two:** [14]

(a) Describe the principle, construction & working of a cyclone.

(b) Explain significance of 'mass balance' in environmental engineering. Also classify waste based on its nature.

(c) What is smog? What are the causes & effects of Smog on human health?

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Useful formulae:

$V = (1/N)\, R^{2/3}\, S^{1/2}$

$h_L = 0.0729(V^2 - v^2)$

$y_t = L_0\,(1 - 10^{-kt})$